國立臺北科技大學
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資料結構
Data Structure

加分題

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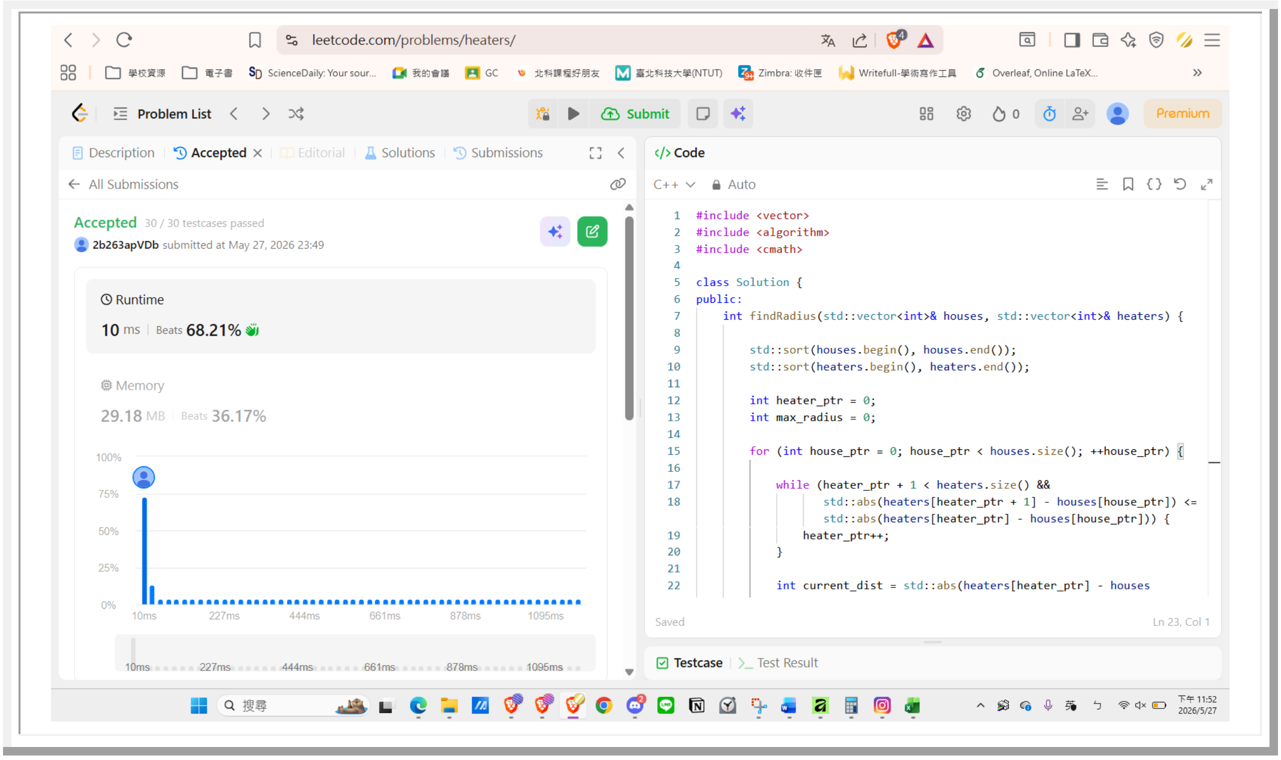
加分題 01-Heater

Given the positions of houses and heaters on a horizontal line, return *the minimum radius standard of heaters so that those heaters could cover all houses.*

Code

```
#include <vector>
#include <algorithm>
#include <cmath>

class Solution {
public:
    int findRadius(std::vector<int>& houses, std::vector<int>& heaters) {

        std::sort(houses.begin(), houses.end());
        std::sort(heaters.begin(), heaters.end());

        int heater_ptr = 0;
        int max_radius = 0;

        for (int house_ptr = 0; house_ptr < houses.size(); ++house_ptr) {

            while (heater_ptr + 1 < heaters.size() &&
                   std::abs(heaters[heater_ptr + 1] - houses[house_ptr]) <=
std::abs(heaters[heater_ptr] - houses[house_ptr])) {
                heater_ptr++;
            }

            int current_dist = std::abs(heaters[heater_ptr] -
houses[house_ptr]);

            max_radius = std::max(max_radius, current_dist);
        }

        return max_radius;
    }
};
```

Result


leetcode.com/problems/heaters/
Problem List
Submit
Premium
Description
Accepted
Editorial
Solutions
Submissions
All Submissions
Accepted 30 / 30 testcases passed
2b263apVDb submitted at May 27, 2026 23:49
Runtime
10 ms | Beats 68.21%
Memory
29.18 MB | Beats 36.17%
Code
C++
Auto
#include <vector>
#include <algorithm>
#include <cmath>
class Solution {
public:
int findRadius(std::vector<int>& houses, std::vector<int>& heaters) {
std::sort(houses.begin(), houses.end());
std::sort(heaters.begin(), heaters.end());
int heater_ptr = 0;
int max_radius = 0;
for (int house_ptr = 0; house_ptr < houses.size(); ++house_ptr) {
while (heater_ptr + 1 < heaters.size() &&
std::abs(heaters[heater_ptr + 1] - houses[house_ptr]) <=
std::abs(heaters[heater_ptr] - houses[house_ptr])) {
heater_ptr++;
}
int current_dist = std::abs(heaters[heater_ptr] - houses
Saved
Ln 23, Col 1
Testcase
Test Result
2026/5/27